

Chase Pellegrom

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Seeking Summer 2026 Internship - Willing to relocate - Open to field and office roles

Education

Colorado School of Mines - Golden, CO

B.S. Mechanical Engineering, Minor in Metallurgical Engineering and Materials Science - Expected May 2028

GPA: 3.7 | Dean's List | Coors Foundation Achievement Scholarship Recipient | Junior Standing

- **Interests:** Fracture mechanics, failure analysis, extreme environmental degradation
- **Organizations:** American Society of Mechanical Engineers (ASME)

Research Experience

Undergraduate Researcher - Colorado Corrosion Center

Colorado School of Mines | Advisor: Dr. Xiolei Guo | December 2025 - Present

Research group website available

- *Durability of Past, Present, and Future Concrete Formulations in Coastal Environments*
 - Performed electrochemical analysis (OCP, LPR, EIS) to quantify steel rebar corrosion rates under simulated coastal salt ponding
- *Corrosion of Additively Manufactured Metallic Alloys*
 - Evaluated crevice corrosion susceptibility of additively manufactured dental alloys under simulated service environments (saliva, fluoride, acidic exposure)

Technical Projects

Sustainable Packaging Redesign

- Redesigned packaging geometry using CAD and basic structural analysis to reduce material usage while maintaining load capacity

Carbon Dioxide Trend Analysis

- Applied Python-based FFT and time-series analysis to isolate long-term and seasonal trends in atmospheric carbon dioxide data

Technical Skills

- **CAD & Modeling:** SOLIDWORKS (CSWA), AutoCAD, Revit, Inventor
- **Machining & Fabrication:** Manual mill, lathe, drill press, band saw, shop safety & measurement
- **Programming & Analysis:** Python, C/C++, Microsoft Office Specialist (MOS), HTML
- **Engineering & Research:** Solid mechanics, structural analysis, experimental design, technical reporting

Experience

Union American Bistro - Castle Rock, CO

Utility | Barback | FOH Support | June 2023 – Present

- Managed time-critical workflows in a high-volume environment
- Coordinated across teams to maintain service efficiency during peak demand